Spring-Mass Harmonic Oscillator

1. Definition

A spring-mass harmonic oscillator is a system in which a mass attached to a spring oscillates about its equilibrium position when displaced and released. For an ideal spring obeying Hooke's law, the motion is simple harmonic motion (SHM).

2. Principle

According to Hooke's law, the restoring force is directly proportional to the displacement and acts in the opposite direction.

$F = -kx$

Where:

- F = restoring force (N)
- k = spring constant (N/m)
- x = displacement from equilibrium (m)

By Newton's second law:

$F = ma$

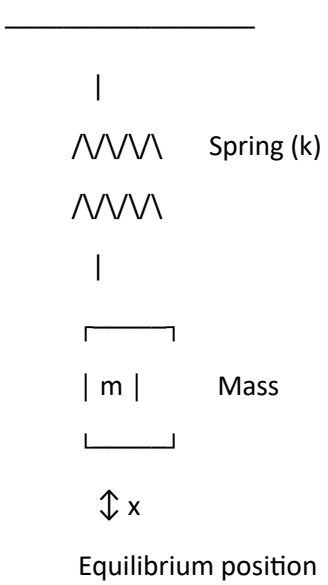
Therefore,

$ma = -kx$

$a = -(k/m)x$

Since acceleration is proportional to displacement and directed towards the equilibrium position, the motion is SHM.

3. Diagram



4. Important Formulae

- Angular frequency: $\omega = \sqrt{(k/m)}$
- Time period: $T = 2\pi\sqrt{(m/k)}$
- Frequency: $f = 1/T = (1/2\pi)\sqrt{(k/m)}$
- Acceleration: $a = -\omega^2 x$
- Displacement: $x = A \sin(\omega t + \phi)$
- Potential energy: $PE = \frac{1}{2}kx^2$
- Total energy: $E = \frac{1}{2}kA^2$

Where:

- m = mass attached to the spring (kg)
- k = spring constant (N/m)
- T = time period (s)
- f = frequency (Hz)
- A = amplitude (m)
- ω = angular frequency (rad/s)

5. Characteristics

1. The mass oscillates about its equilibrium position.
2. The restoring force always acts towards the equilibrium position.
3. The time period increases when the attached mass increases.
4. The time period decreases when the spring constant increases.
5. For an ideal spring, the time period is independent of amplitude.
6. At the equilibrium position, speed is maximum and acceleration is zero.
7. At the extreme positions, speed is zero and acceleration is maximum.

6. Energy in SHM

- At the equilibrium position: kinetic energy is maximum and spring potential energy is minimum.
- At the extreme positions: kinetic energy is zero and spring potential energy is maximum.
- In an ideal system without friction, total mechanical energy remains constant.

7. Result

A mass attached to an ideal spring performs simple harmonic motion when displaced slightly from its equilibrium position and released. Its time period is:

$T = 2\pi\sqrt{(m/k)}$

The time period depends on the attached mass and the spring constant.